

Mid-project progress report

Work was completed this period:

In the current period, we had to perform tasks associated with planning and implementation phase of development of our Clinic Database System.

In the first place, we had to consider requirements of the project and get acquainted with the information stored in the database. Based on the scenario, there are 13 entities.

Secondly, we had to identify attributes and primary and foreign keys of all tables. In addition, it should be noted that great attention was paid to the relationships between the entities in order to provide the integrity and normalization of the database. Also, we have stated all assumptions that were taken into account in the process of developing the database. Another important activity is creation of the EER diagram that presents all relationships between the entities and specialization hierarchy between the supertype Entity and the subtypes. After the completion of all tasks of the database design, we implemented the database system using MySQL. In particular, we have created our Clinic database and described all tables using SQL statements. We have identified the data types and constraints, including such constraints as PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE and CHECK constraints.

Finally, we have added some test records into all main tables and verified whether the relationships between the tables function correctly. The testing of the database helped to detect and fix some errors prior to using SQL statements. To summarize, Tasks 1 and 2 were performed successfully and laid the groundwork for the further phases of the project implementation and testing.

One key decision made during this period:

Among the decisions taken by our group was the use of the EER specialization technique while creating the Employee entity.

Although we had the option to create individual entities of Doctor, Nurse, Pharmacist and Staff containing all the relevant information concerning the employees, we decided to use the Employee as the supertype and Doctor, Nurse, Pharmacist and Staff as the subtypes.

This decision allowed us to eliminate redundancy to a great extent since all the redundant information concerning the employees such as name, telephone, e-mail, salary and date of hiring would have to be entered only once.

One problem encountered:

The first challenge that we encountered was ensuring that the tables are created in the correct order and functioning properly. Given the large number of tables with foreign key constraints, any attempt to create a child table prior to its parent table will trigger SQL errors.

In our case, for example, the Appointment table depends on the Patient and the Doctor table and the Doctor table depends on the Employee table. In other words, there was a need to give some time to create an appropriate order in which the tables will be created such that each referenced table will be created before its child table.

The other challenge we encountered was the preparation of realistic data. Since all foreign keys refer to existing records, it was necessary to create patients, departments, employees, and doctors before creating appointments, treatments, prescriptions, and payments as possible.

Plan to do next:

We are planning to do Task 3 and Task 4, that they contain these subtasks:

- Writing SELECT queries.
- Writing JOIN queries.
- Writing nested queries.
- Writing of aggregate queries with GROUP BY.
- Writing UPDATE and DELETE statements.
- Construction of one VIEW.
- Construction of one TRIGGER.
- Testing of all necessary SQL statements with screen captures of SQL queries and the results.
- Check all the database to ensure that all the constraints work properly.
- Uploading the final SQL script to the project's GitHub repository for version control and submission.

Each team member input:

Team member	Their input
Nawaf Allahyani	Check the requirements of the project. Helped in defining database entities, attributes and relationships. Checked the ER diagram.
Abdullellah Khalid Alali	Helped prepare the documentation and assumptions. Tested the SQL implementation. Verified the relationships between tables.
MOHAMMED ALSHAMMASI	Database design assisted. Primary and foreign keys defined. Normalization checked, and ER diagram prepared.
Abdulwahid Alqatari	Implemented the database. Created all SQL tables, inserted the sample data, tested the database implementation. Will upload the final SQL script to the project's GitHub repository.